

Contribution-Aware Trust Mechanism for Poisoning Attack Detection in Swarm Learning

Architecture Components Explained

1. Overview

In Swarm Learning (SL), many participants (peers) train a shared model together without sending their private data anywhere and without a central server. The risk is that one peer may be malicious and send bad updates (a poisoning attack) to damage the shared model. This architecture adds a Trust and Defense Layer that checks every peer's update, gives each peer a trust score, and lets trusted peers influence the shared model more than suspicious ones.

2. Architecture Diagram

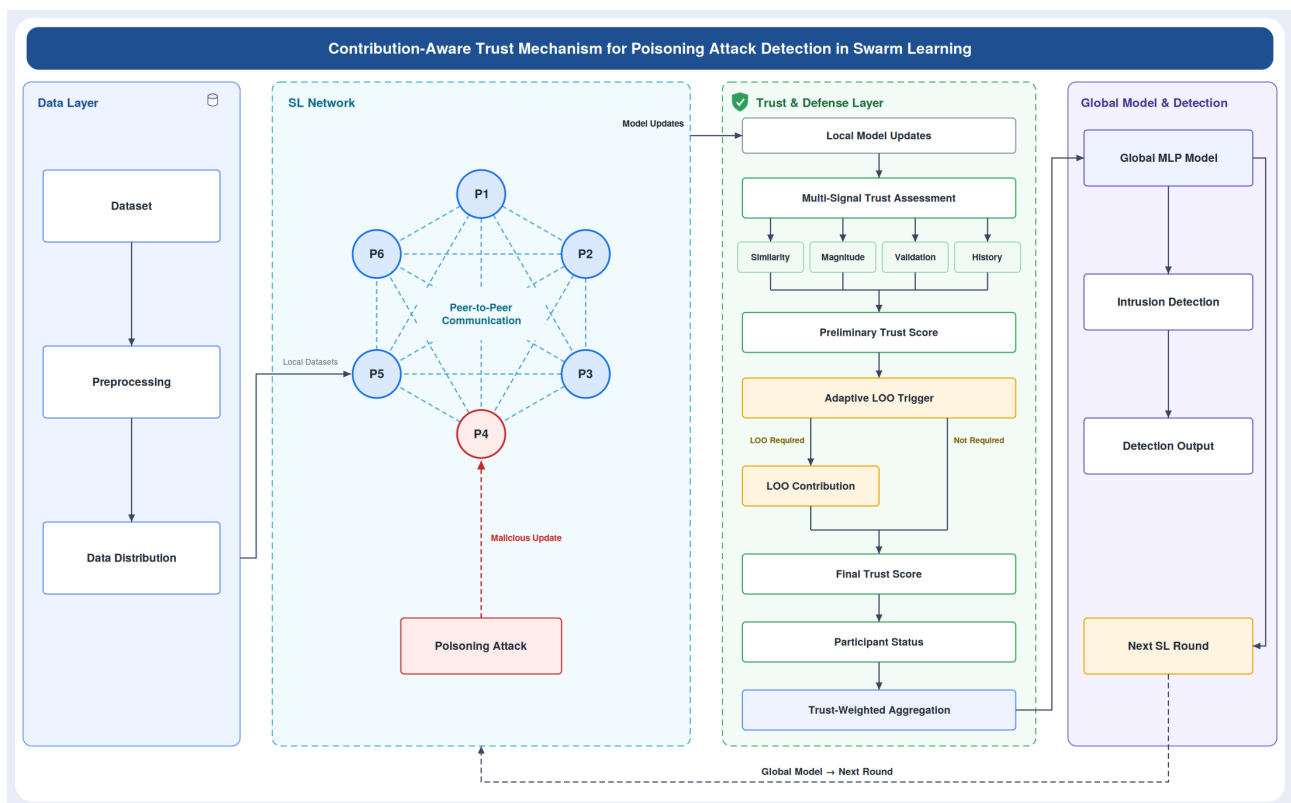


Figure 1: High-level system architecture

3. Components

The architecture has four parts, read left to right in the diagram.

3.1 Data Layer

Dataset	The raw data used to train the model. It is the starting point of the whole system.
Preprocessing	Cleans and prepares the raw data so the model can learn from it (for example, converting values into a common format and scale).
Data Distribution	Splits the prepared data among the peers. In the IID setting every peer gets a similar mix of data. In the Non-IID setting each peer gets a different mix, which is closer to real life and makes honest peers look different from each other.

3.2 SL Network

Peers (P1 to P6)	The participants of the swarm. Each peer keeps its own data, trains a model on it locally, and shares only the resulting model update, never the data.
Peer-to-Peer Communication	Peers are connected directly to each other. There is no central server, so no single point of failure or single point of control.
Poisoning Attack	A malicious peer (shown in red) sends a harmful update, such as one that is flipped, scaled up or noisy, to damage the shared model. This is the threat the system is designed to handle.

3.3 Trust and Defense Layer

This is the core of the project. It receives the peers' updates and decides how much each one should count. The steps run from top to bottom.

Local Model Updates	The updates sent by all peers in the current round. These are the input to the defense.
Multi-Signal Trust Assessment	Checks every update using four different signals instead of just one, so that no single weakness can be exploited.
Similarity	Does the update point in roughly the same direction as the other peers' updates? Updates pointing the opposite way are suspicious.
Magnitude	Is the update a normal size? Extremely large updates are a sign of an attempt to dominate the shared model.
Validation	Does applying the update improve or hurt accuracy on a small validation set? Updates that hurt accuracy are suspicious.
History	How has this peer behaved in earlier rounds? A peer with a good record is not punished for one odd update, and a repeat offender loses trust.
Preliminary Trust Score	The four signals are combined into a first trust score for each peer.

Adaptive LOO Trigger	Decides whether a deeper check is needed this round. Leave-One-Out (LOO) is expensive, so it runs only when needed, for example on a regular schedule, when accuracy drops, or when a peer looks unusual.
LOO Contribution	Leave-One-Out: the model is evaluated with and without a peer. If the model gets better without that peer, the peer is harmful. If it gets worse, the peer is useful. This protects honest peers whose data is just different.
Final Trust Score	The preliminary score, adjusted by the LOO result when LOO was run. When LOO was skipped, the preliminary score is used as it is.
Participant Status	Based on the final score, each peer is labelled: Trusted, Normal, Suspicious or High Risk.
Trust-Weighted Aggregation	The updates are merged into one model, with each peer's influence proportional to its trust. Low-trust peers are down-weighted, not thrown out, so useful information from honest but unusual peers is kept.

3.4 Global Model and Detection

Global MLP Model	The shared model (a Multi-Layer Perceptron neural network) produced by the trust-weighted aggregation.
Intrusion Detection	The global model is applied to new incoming data to decide whether it is normal or malicious.
Detection Output	The final result of the system: the prediction for each input.
Next SL Round	The updated global model is sent back to all peers, who train again. The dashed arrow in the diagram shows this loop, which repeats every round.

4. How It Works, Step by Step

1. The dataset is prepared and divided among the peers (IID or Non-IID).
2. Each peer trains locally and sends only its model update to the swarm.
3. A malicious peer may send a poisoned update.
4. The Trust and Defense Layer scores every update using Similarity, Magnitude, Validation and History.
5. If needed, LOO checks whether each peer helps or harms the model.
6. Each peer receives a final trust score and a status.
7. Updates are combined with trust-based weights to form the global model.
8. The global model is used for detection and sent back for the next round.

5. Key Idea

Different does not mean malicious. Honest peers with unusual data can look different from the rest. By combining several signals with a Leave-One-Out check that measures real usefulness, the system can catch harmful peers while still keeping honest ones.